Seed2Ontology Enterprise Platform

World-Class Architecture · CMMI Level 5 Compliant · LLM-Driven · Standards-Aligned · Grounded in Real-World Data · Governed End-to-End

CMMI Level 5 | 38 Enterprise Modules | 7 Pipeline Layers | 14+ Global Standards | 5-Tier Validation | 100% Provenance | Continuous Feedback | CQ-Tested | Human-in-the-Loop | RBAC & Audit | Data Governance | Observability | Enterprise Analytics | Managed Lifecycle

Standards Mapping Catalog (M10)

Generic Vocabularies
- Schema.org · Dublin Core (DCAT, DCTERMS)
- W3C SKOS · PROV-O (Provenance)

Geospatial
- OGC GeoSPARQL · W3C WGS84

IoT / Sensors
- W3C SSN / SOSA · SAREF

Financial
- FIBO (Financial Industry Business Ontology)

Top-Level Ontologies (ISO-Aligned)
- BFO · DOLCE
- ISO/IEC 21838 (Top-Level Ontology)
- ISO 29002 (Concept Dictionary ↱ Ontology)

Domain Bridges (Examples)
- Financial ↱ FIBO
- Geospatial ↱ GeoSPARQL + WGS84
- IoT ↱ SOSA / SSN / SAREF
- Climate Risk ↱ Multi-domain crosswalk
- Healthcare ↱ SNOMED CT / UMLS

Service Properties
- Versioned alignment files (SemVer)
- Reusable across projects · drift detection
- Coverage & conformance reports per release

Consumption
- Used by M06 (Upper-Onto Alignment)
- Used by M09 (Standards Conformance)
- Used by M16 (Data Quality Framework)

Data Foundation Modules (M15–M20)

M15 Data Ingestion Framework
Unified connectors: PDF, JSON, XML, CSV, SQL, KG · Multimodal parsers (LlamaParse, Unstructured.io) · Real-time & batch modes
Schema introspection · Provenance capture per chunk
Connector marketplace · SLA-driven retry & backoff
Feeds M20 Metadata & Ontology Service

M16 Data Quality Framework
6 quality dimensions: completeness, accuracy, consistency, timeliness, validity, uniqueness · Automated DQ scorecards
Rule engine (Great Expectations-compatible) · Anomaly detection
Data profiling · Threshold-based quarantine & alerts
DQ gates before Layer 2 extraction

M17 Data Dictionary Service
Business glossary with term definitions, synonyms, ownership
Technical schema catalog (tables, columns, types, constraints)
Lineage links: source ↱ transformation ↱ ontology
Stewardship assignments · Definition approval workflow
Feeds M05 disambiguation · Prevents semantic drift

M18 Data Lifecycle Management
Retention policies · Archival · Purge (with audit trail)
Tiered storage: hot ↱ warm ↱ cold · Cost optimization
Legal hold · Right-to-be-forgotten workflows (GDPR/CCPA)
Data versioning & lineage preservation across deletions
Compliance reporting · Records management integration

M19 Data Governance Center
Data ownership & stewardship · Policies & standards
Classification (PII, confidential, public) · Access reviews
Policy-as-code engine · Compliance dashboards
Lineage governance · Change approval workflows
Audit trail integration with M27 Provenance

M20 Metadata & Ontology Service
Central metadata catalog: technical + business + operational
Ontology artifact registry (TBox, ABox, shapes) — versioned
Search & discovery · Tag/annotation management
Bidirectional links to M17 Data Dictionary and M28 Registry
API: REST + GraphQL + SPARQL · RBAC-scoped visibility

Data Foundation — Value Chain
Data ingestion (M15) ↱ Quality gates (M16) ↱ Dictionary registration (M17) ↱ Lifecycle (M18) ↱ Governance policy application (M19) · ↱ Metadata & ontology publication (M20)

Data Foundation — KPIs
- Ingestion success rate ≥ 99.5%
- DQ score ≥ 95% for production sources
- Dictionary coverage ≥ 90% of curated terms
- Retention policy compliance = 100%
- Metadata freshness ≤ 15 min

Governance Alignment
- CMMI-5 MA, CM, PPQA integration
- ISO 27001, SOC 2, GDPR/CCPA readiness

Feedback Loop

Layer 0 · Seed Formalization & Competency-Question Studio (M01)

Seed Contract
Competency Questions · Scope boundaries · Vocabulary constraints · Acceptance criteria
Intended reasoning tasks · Target simulation · Fixed upper-ontology commitment (BFO/DOLCE)
Human approval required before pipeline execution

Versioned Seed Registry
Immutable seed snapshots · Change log · Approval workflow · Human sign-off
Linked to project, domain, regulatory context · Feeds Evaluation Harness (M29)
Referenced by all downstream layers via immutable seed ID

Layer 1 · Data Ingestion Framework & Contextual Analysis (M15 · M02)

Multimodal Ingestion (M15)
PDFs · Reports · JSON · XML · CSV · SQL · KG
Parsers: LlamaParse, Unstructured.io
OCR · Table extraction · Metadata capture
Provenance per chunk · Real-time + batch
Connector marketplace · Retry & backoff
Rejected chunks ↱ DQ quarantine (M16)

Simulation Requirement Analyzer (M02)
Boundaries · Agents · Variables · Constraints
KPIs · Time horizon · Digital-twin vs. ABM
Elicits implicit requirements via LLM
Drafts CQs ↱ sent to Layer 0 for review
Links simulation intent to ontology scope
Reviewed by domain SMEs (M12)

Metadata & Schema Profiling
Schema introspection · API payload sampling
Table/column profiling · Range & distribution stats
Feeds M16 Data Quality, M20 Metadata, M17 Dictionary
Provenance per source artifact
CQ elicitation drafts ↱ Layer 0
Emits data lineage edges to M19 Governance

Layer 2 · Real-World Seed Extraction (M03 · M04 · M05)

Entity & Relation Mining (M03)
NER · RE · GLiNER · LLM prompts
Candidate classes, properties, instances
Provenance + confidence per candidate
Evidence spans preserved · Human spot-check via M12
Uses M17 Dictionary for canonical terms
DQ-scored inputs from M16

Statistical Grounding (M04)
Cross-reference with DB columns
API payloads · Schemas · Data ranges
Rejects hallucinated elements
Coverage metric per seed · Evidence count
Uses M16 DQ profiles · M20 metadata
Candidate rejected ↱ logged to M27 Provenance

Disambiguation & Clustering (M05)
Embedding similarity · HDBSCAN · FAISS/Qdrant
Canonical term resolution against M17 Dictionary
Synonym merge proposals · Human-approved
Versioned seed output
Drift alerts to M21 Data Observability
Approved merges committed to M20

Layer 3 · LLM-Driven Ontology Synthesis (M06 · M07 · M08)

Upper Ontology Alignment (M06)
Fixed commitment: BFO or DOLCE per project
Explicit mapping rules · Endurant vs. Perdurant checks
Violation detection · Review via M12 Console
Human sign-off for high-impact mappings
Uses M10 Standards Mapping Catalog
Decision chain ↱ M27 Provenance

Domain Ontology Synthesizer (M07)
Subclasses · Object & data properties
Restrictions · Axioms · Disjointness
Decision chain per axiom (prompt, model version)
Structured diff ↱ human review (M12)
Uses M17 Dictionary terms · M10 standards
Committed to M28 Registry after validation

Serialization & Schema Export (M08)
Turtle (.ttl) · RDF/XML · OWL Manchester
SHACL shapes (draft) · ShEx (draft)
Namespace & IRI conventions
Committed to M28 Registry (versioned)
Emits to M20 Metadata Service for search
Feeds M09 Validation Orchestrator

Layer 4 · Standardization & Multi-Tier Validation (M09 · M10 · M12)

Syntactic (M09)
RDF/XML well-formed
Turtle well-formed
Namespace checks
RDFLib / Jena parse
Errors ↱ M27 Provenance
CMMI-5: PPQA

Structural (M09)
SHACL / ShEx validation
Cardinality · Datatype
Class disjointness
Domain / range checks
pyshacl / ShEx.js
CMMI-5: PPQA · VER

Logical (M09)
HermiT / Pellet / ELK
Owlready2 consistency
Unsatisfiable classes
Disjointness violations
Explains / justification
CMMI-5: PPQA · VAL

Quality (M09)
OOPS! / OntoClean
Pitfall detection
Naming conventions
Rigidity / identity
Annotation completeness
CMMI-5: PPQA

Standards Conformance (M10)
Schema.org · Dublin Core
SKOS · PROV-O
FIBO · SOSA/SSN · SAREF
GeoSPARQL · WGS84
BFO/DOLCE · ISO 21838/29002
CMMI-5: DAR · VER

Human Review Gate (M12)
Triage by severity · Approve / reject / edit · Dual review for schema-changing decisions · Decision recorded in M27 · Gate before ABox (M11) and Simulation (M13)
Escalation SLAs · SME sign-off · Compliance evidence auto-captured · Feeds M22 Success Metrics and M23 KPI Metrics

Layer 4.5 · ABox Population & SPARQL Competency-Question Testing (M11)

Instance Extraction & Linking
ABox entities linked to TBox classes
Entity resolution across sources
Provenance per instance · Link prediction (LLM-assisted)
Confidence scoring · Human review via M12
CMMI-5: PPQA · VER

SPARQL CQ Tests
Every CQ from Layer 0 executed
Pass / fail / partial results
Query templates registered per CQ
Results ↱ M29 Evaluation Harness
CMMI-5: VER · VAL · MA

Coverage & Gap Report
Per-CQ coverage %
Missing classes / properties
Suggested patch seeds ↱ Layer 5
Trendline across versions · Published to M24 Analytics
CMMI-5: MA · QPM

Layer 5 · Simulation Feedback, Patch Governance & Evolution (M13 · M14)

Deployment & Simulation (M13)
Digital twin runtime · Agent-based model
Hybrid simulation · Neo4j/GraphDB
SPARQL endpoint exposed
Version pinned at deploy time · Rollback ready
RBAC-scoped access via M26
Deployment events ↱ M23 KPI framework
CMMI-5: CM · PI
CM audit trail on every deploy

Error Monitoring
Missing-link alerts · Rigid-structure failures
CQ execution regressions
Simulation KPI drift
Agent decision failures
Feedback ↱ M11 gap report
Threshold escalation
CMMI-5: PMC · CAR
Root cause ↱ M34 CAR Service

LLM Patch Proposals (M14)
Agent analysis error logs
Proposes new properties, axioms
Relaxation of constraints
Impact analysis vs. active sims
Draft patch stored in M28 Registry
Never auto-applied · Human approval required
CMMI-5: DAR · CAR
Decision rationale captured in M27

Patch Governance (M14)
Proposal ↱ Impact Analysis ↱ Regression Test Suite ↱ Human Approval (M12) ↱ Versioned Release (M28) ↱ Deploy (opt-in) ↱ Rollback available
CMMI-5: CM · DAR · PPQA
Full audit trail per patch

Continuous Learning Loop
Simulation errors ↱ patch proposals ↱ human review ↱ versioned ontology ↱ re-deploy · regression-tested · fully audited · KPI-tracked via M23 · CQ-tested via M11

Enterprise Consumers & Integration

Agentic AI Runtime
SPARQL / GraphQL query interface
Ontology-driven tool selection
Policy-aware action planning
CQ-linked capability discovery
RBAC-scoped access via M26
Usage analytics ↱ M25

Digital Twin Platform
Semantic simulation engine
Operational semantics via ontology
Missing-link alerts ↱ Layer 5
KPI drift monitoring
Deployment via M13
Feedback loop integration

BI & Analytics Consumers
Self-service BI (M24)
Executive dashboards (M22)
Operational KPIs (M23)
Adoption analytics (M25)
Data observability views (M21)
Governance reports (M19)

External APIs
SPARQL 1.1 endpoint
GraphQL API
REST / OpenAPI
Webhooks · Kafka events
Standards-conformant exports
RBAC enforced at gateway

Analytics, Observability & Measurement (M21–M25)

M21 Data Observability Platform
Pipeline health · Data drift · Volume anomalies · Quality signals · Lineage monitoring · Alerting
Feeds M23 KPI · Integrates M16 DQ · Emits to M24

M22 Success Metrics Framework
Business KPI definition (OKRs, goals, targets) · Metrics taxonomy · Attribution to pipeline modules
Roll-up to executive dashboard via M24

M23 KPI Metrics Framework
Operational measurement infrastructure: SLIs, SLOs, error budgets · Metric storage & querying
Real-time computation · Historical trending · CMMI-5 MA alignment

M24 Analytics Platform
Self-service BI · Dashboards · Ad-hoc query · Report builder · Embedded analytics · Data products
Consumes M22/M23 metrics · Publishes to consumers · RBAC-scoped

M25 Visitor & Usage Analytics
User behavior · Feature adoption · Session analytics · Funnel tracking · Privacy-first (consent-based)
Adoption KPIs feed M22 Success Metrics · GDPR-compliant by design

Enterprise Foundation Services (M26–M31)

M26 RBAC & Identity Access Management
OAuth2 / OIDC · Roles, groups, permissions · Row/column/attribute-level access
Granular ACLs on ontology artifacts, data sources, dashboards
Just-in-time access · Approval workflows · Sessions & MFA

M27 Provenance & Audit Service
PROV-O–aligned decision chain for every class, property, axiom, patch
Records source, prompt, LLM version, reviewer, result · Immutable event log
Query API · Audit reports · Compliance evidence packages

M28 Ontology Registry & Versioning
Semantic versioning (SemVer) · Diff · Rollback · Dependency graph
Immutable releases · Git LFS / DVC integration · Metadata publication to M20
Deployment gate for M13 · Aligns with CMMI-5 CM

M29 Evaluation Harness
Continuous scoring across 4 families: ontology quality, standards alignment, simulation impact, human acceptance · Trend across releases
Composite KPI dashboard · CMMI-5 MA + QPM integration

M30 Security & Compliance Center
PII redaction (Presidio) · Secrets management (Vault) · Encryption at rest/transit
Compliance frameworks: SOC 2, ISO 27001, GDPR, HIPAA
Audit logging · Incident response integration · Posture dashboards

M31 LLM Agent Orchestrator
Prompt versioning · Retries · Guardrails · Cost controls · LangGraph/Prefect
Multi-agent coordination · Model routing · Evaluation integration (LangSmith)
Observability & prompt drift detection · Failover across providers

CMMI Level 5 Process Areas (M32–M38)

M32 Organizational Process Performance (OPP)
Baseline performance metrics across projects · Process capability baselines
Quality & process performance models (PPMs)
Establishes quantitative targets per project based on baselines
CMMI-5 Level 4 PA — feeds M33 QPM

M33 Quantitative Project Management (QPM)
Statistical process control (SPC) on KPI streams · Control charts
Quantitative project objectives management
Sub-process performance analysis · Predictability measurement
CMMI-5 Level 4 PA — integrates M23 KPI, M29 Evaluation

M34 Causal Analysis & Resolution (CAR)
Root cause analysis on defects, regressions, incidents
5 Whys · Fishbone · Fault tree · Causal graphs
Action tracking · Preventive change management
CMMI-5 Level 5 PA — continuous improvement driver

M35 Decision Analysis & Resolution (DAR)
Formal evaluation of alternatives (patch choices, standards, LLM models)
Decision criteria weighting · Trade-off documentation
Rationale recorded in M27 Provenance
CMMI-5 Level 3 PA — applied to high-impact decisions

M36 Configuration Management (CM)
Baseline identification · Configuration control · Status accounting
Configuration audits · Applies to all pipeline artifacts
CMMI-5 Level 2 PA — integration with M28 Registry

M37 Process & Product Quality Assurance (PPQA)
Objective evaluation of processes & work products against standards
Non-compliance tracking · Escalation procedures
CMMI-5 Level 2 PA — covers validation orchestration, review gates

M38 Measurement & Analysis (MA)
Measurement objectives · Metric specification & data collection
Measurement analysis · Results communication
CMMI-5 Level 2 PA — foundational for OPP, QPM, CAR

CMMI Level 5 — Continuous Representation
Process areas integrate with pipeline (M01–M25): OPP/QPM consume KPI metrics · CAR consumes incident streams · DAR consumes decisions · CM covers all artifacts · PPQA gates validation · MA feeds OPP.
CMMI-5 audit evidence auto-collected across all 38 modules.

Legend: Seed | Ingest | Extract | Synthesize | Validate | ABox/CQ | Evolve | Data Foundation | Analytics & Observability | Enterprise Foundation | CMMI-5 Process Areas

Architecture: 7 pipeline layers · 6 data foundation modules · 5 analytics modules · 6 enterprise foundation services · 7 CMMI-5 process areas · 38 total modules

Standards: BFO · DOLCE · ISO/IEC 21838 · ISO 29002 · Schema.org · DCAT · DCTERMS · SKOS · PROV-O · GeoSPARQL · WGS84 · SSN · SOSA · SAREF · FIBO

Compliance: CMMI Level 5 (Continuous Representation) · SOC 2 · ISO 27001 · GDPR · HIPAA · SPARQL 1.1 · OWL 2 · SHACL · ShEx

Seed2Ontology Enterprise Platform v2.0 — Regenerated Reference Architecture

38 modules · 7 pipeline layers · CMMI Level 5 compliant · Production-grade